BUSINESS REQUIREMENTS DOCUMENT

BRD Hospital — Gynaecology, Pregnancy & Maternity Care Platform

Marketing Website · Online Appointment Booking · Admin Dashboard

Document Type	Business Requirements Document (BRD)
Project	BRD Hospital — Gynaecologist & Maternity Care Website
Prepared By	Craftlanee (Development Partner)
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Document Control

Revision History

Version	Date	Author	Description
0.1	01 Sep 2026	Craftlanee	Initial draft based on delivered build
1.0	10 Sep 2026	Craftlanee	Finalized BRD covering full feature set, tech stack, and data model

Approval

Role	Name	Signature	Date
Project Sponsor / Client	BRD Hospital (Dr. Haritha Mandava)		
Development Partner	Craftlanee		

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1. Executive Summary

BRD Hospital is a digital front door for a gynaecology and maternity care practice led by Dr. Haritha Mandava. This document defines the business and functional requirements for the platform that has been designed and built: a patient-facing marketing website, an online appointment booking system, and a private admin dashboard for staff to manage bookings, availability, and payments.

The platform is built as a modern, fast, mobile-responsive single-page application, backed by a managed cloud database, and is already deployed and operational. This BRD captures what the system does, why it does it, the technology used to build it, and the data and workflows that power it — serving as the single reference document for stakeholders, new team members, and future enhancement planning.

The patient experience the platform is designed to reflect — a calm, trusted maternity care journey.

2. Project Background & Purpose

Expecting mothers and women seeking gynaecological care need clear information, an easy way to book a consultation, and confidence that they are choosing a qualified, experienced doctor. Historically, this kind of practice relied on phone calls and walk-ins, with no online presence to build trust before a patient's first visit, and no digital record of bookings for staff to manage.

The purpose of this project is to give BRD Hospital a professional online presence that:

- Builds trust with prospective patients through the doctor's credentials, real photography, and a warm, emotionally resonant design.
- Lets a patient book an appointment online in a few steps, at any time, without a phone call.
- Gives clinic staff one place to see, confirm, and manage every booking, its payment status, and doctor availability.

- Works for patients across language backgrounds common in the region (English, Hindi, Telugu, Tamil, Kannada).

3. Business Objectives

#	Objective	Success Measure
1	Establish a credible, trustworthy online presence for the practice	Professional marketing site live and accessible 24/7
2	Reduce phone-based booking overhead	Patients can self-book a consultation online in under 2 minutes
3	Give staff a single source of truth for appointments	All bookings visible and manageable from the admin dashboard
4	Support informal, low-friction digital payments	UPI QR + WhatsApp confirmation flow used at booking
5	Serve a multilingual patient base	Site available in 5 languages
6	Keep the site fast and reliable on mobile devices	Optimized images, responsive layout, sub-second navigation

4. Scope of Work

4.1 In Scope

- Public marketing website covering the doctor's profile, services, patient journey, and testimonial-style content.
- Multi-step online appointment booking flow with live availability checking.
- Private, authenticated admin dashboard for staff (appointments, availability, payments, settings).
- Manual UPI-based payment flow with WhatsApp-assisted confirmation (no payment gateway integration).
- Automated email notifications to staff on new bookings and to patients on confirmation.
- Multilingual UI (English, Hindi, Telugu, Tamil, Kannada).
- Cloud hosting and deployment pipeline for the live site.

4.2 Out of Scope

- Direct online payment gateway (card/UPI auto-capture) integration.
- Multi-doctor / multi-branch scheduling (current build supports a single doctor).
- Native mobile applications (iOS/Android) — the site is a responsive web app only.
- Electronic Health Records (EHR) or clinical charting functionality.
- Third-party marketing automation (SMS campaigns, CRM integration).

5. Stakeholders

Stakeholder	Role / Interest
Dr. Haritha Mandava	Practicing obstetrician, gynaecologist & sonologist; primary practitioner represented on the site; approves clinical/content accuracy
Clinic Front-Desk / Admin Staff	Daily users of the admin dashboard — manage appointments, availability, and payment tracking
Patients / Site Visitors	End users of the public site and booking flow
Craftlanee (Development Partner)	Designs, builds, deploys, and maintains the platform

Dr. Haritha Mandava — Obstetrician, Gynaecologist & Sonologist, the practice's primary practitioner.

6. Technology Stack

The platform is built entirely on a modern JavaScript/TypeScript stack with a serverless backend — there is no separate application server to provision or maintain.

6.1 Frontend

Technology	Purpose
React 19	Core UI library — component-based single-page application
TypeScript	Type-safe application code across the entire frontend
Vite	Build tool and local dev server — fast hot-reload and optimized production bundles
React Router DOM	Client-side routing (marketing site, booking, admin routes)
Tailwind CSS v4	Utility-first styling framework for the entire design system
Framer Motion	Animations and scroll-based reveal effects throughout the site
Lucide React	Icon library used across the UI
Custom i18n context	Lightweight in-house translation system (not a third-party i18n library)

6.2 Backend & Data

Technology	Purpose
Supabase (PostgreSQL)	Managed relational database holding doctors, services, appointments, availability, and settings
Supabase Auth	Email/password authentication securing the admin dashboard
Supabase Edge Functions (Deno)	Serverless functions for outbound email — new-booking staff alerts and patient confirmations
Row-Level Security (RLS)	Postgres policies restricting data access by role, enforced at the database layer
@supabase/supabase-js	JavaScript client the frontend uses to talk to the database and auth directly (no custom API server)

6.3 Supporting Libraries & Tooling

Technology	Purpose
qrcode	Generates the UPI payment QR code shown during checkout
sharp	Server-side image optimization during the build/asset pipeline
Playwright	Used ad hoc for automated image-processing scripts
oxlint	Fast linter enforcing code quality and consistency
PostCSS + Autoprefixer	CSS processing pipeline supporting Tailwind
npm	Package management and build scripts

6.4 Hosting & Deployment

Technology	Purpose
Netlify	Production hosting, CDN, and CI/CD — auto-deploys on every push to the main branch
Git / GitHub	Source control and deployment trigger
Unsplash CDN	Supplemental stock photography served alongside the clinic's own photos

7. System Architecture

The application follows a modern JAMstack-style architecture: a statically-built, client-side React application is served from Netlify's global CDN, and talks directly to Supabase (hosted PostgreSQL + Auth + Edge Functions) over HTTPS for all dynamic data. There is no custom backend server to deploy, patch, or scale — Supabase and Netlify handle that layer.

Request Flow

- A visitor's browser loads the pre-built React app (HTML/CSS/JS) from Netlify's CDN.
- Public content (services, doctor profile, journey content) is bundled with the app or loaded from local/Unsplash image sources — no database call required to browse the marketing site.
- Booking a slot calls Supabase directly (via supabase-js) to check availability and insert an appointment row, protected by Row-Level Security.
- A successful booking triggers the notify-new-booking Edge Function, emailing staff immediately.
- When staff update a booking's status/payment in the admin dashboard, the send-confirmation Edge Function emails the patient.
- Admin routes are gated by Supabase Auth; unauthenticated visitors are redirected to /admin/login.

Without Supabase configured (e.g. in a fresh local environment), the marketing site still renders fully — booking and admin features degrade gracefully instead of crashing the app, controlled by an isSupabaseConfigured flag.

8. Functional Requirements

8.1 Public Marketing Website

The homepage is assembled from focused, single-purpose sections designed to build trust and guide the visitor toward booking:

- **— introduces the practice with a warm, high-impact visual and a primary call-to-action to book.**Hero
- **— quick-glance credibility markers (experience, patients cared for, ratings).**Trust Strip
- **— Dr. Haritha Mandava's qualifications, specializations, languages spoken, and availability.**Doctor Spotlight
- **— content mapped to each trimester through delivery and postnatal care.**Pregnancy Journey
- **— an emotionally-driven storytelling section (crying, laughing, cooing moments) reinforcing the human side of care, each with supporting audio.**Baby Emotions
- **— photography-led sections building emotional connection.**Mother & Baby Story / Emotional Collage
- **— key numbers (years of experience, patients served, satisfaction).**Stats
- **— a closing prompt to book an appointment.**Final Call-to-Action
- **— instantly re-renders all UI text in the visitor's preferred language.**Language Switcher

Hero storytelling imagery used across the marketing site's pregnancy-journey sections.

8.2 Online Appointment Booking

A guided, multi-step booking wizard lets a patient reserve a consultation without calling the clinic:

- **(currently: Gynecological Consultation, 30 minutes).**Step 1 — Service selection
- **via a calendar date picker; already-booked and blocked slots are automatically disabled.**Step 2 — Date & time selection
- **— full name, phone, email, date of birth, reason for visit (from a configurable checklist), and an optional message.**Step 3 — Patient details
- **— a UPI QR code is displayed with the configured payee details; the patient sends payment via any UPI app, optionally messages a screenshot over WhatsApp, and enters their UPI transaction ID.**Step 4 — Payment
- **— the booking is saved to the database and staff are notified by email immediately.**Confirmation

Availability is layered: a slot is unbookable if it is already booked, individually blocked by staff, or falls on a day staff have blocked entirely — the calendar greys out fully-blocked days automatically.

8.3 Admin Dashboard

A private, authenticated dashboard (/admin) gives staff full operational control:

Tab	Capability
Overview	At-a-glance summary of bookings and activity
Appointments	View every booking; update status (pending / confirmed / cancelled / completed / no-show) and payment (pending / paid / failed) independently; search by name, phone, email, or UPI transaction ID; filter by date; open full patient detail
Availability	Block an individual time slot, or block an entire day, on top of the fixed weekly schedule
Payments	Track manually-verified payment status per booking; export records to CSV for offline records
Settings	Configure booking fee, UPI ID, payee name, WhatsApp contact number, and the reason-for-visit checklist

- **— plain email/password via Supabase Auth; staff accounts are created manually (no public self-signup), by design.**Authentication
- **— appointments are fetched once per session and shared across tabs for consistency and performance.**Shared data loading

8.4 Notifications

- **— staff receive an email the moment a patient completes a booking (notify-new-booking Edge Function).**New-booking alert
- **— the patient receives a confirmation email once staff mark the booking's status and payment (send-confirmation Edge Function).**Patient confirmation
- **— a click-to-chat WhatsApp link is offered during payment for the patient to share a payment screenshot directly with the clinic.**WhatsApp assist

8.5 Multilingual Support (i18n)

Every user-facing string is centralized and keyed by section, with a lightweight custom translation context (not a third-party i18n library) exposing t() for single strings and tList() for lists (e.g. checklists). The site currently supports:

Language	Code
English	en
Hindi	hi
Telugu	te
Tamil	ta
Kannada	kn

9. Non-Functional Requirements

Category	Requirement
Performance	Optimized (webp/avif) and lazily-loaded imagery; Vite-bundled production build; content served from Netlify's global CDN
Responsiveness	Fully responsive layout across mobile, tablet, and desktop viewports (Tailwind CSS)
Availability	Static frontend on Netlify CDN plus managed Supabase infrastructure — no single application server to fail
Security	Row-Level Security on all Supabase tables; admin routes gated by authenticated sessions; no public self-signup for staff accounts
Graceful degradation	The marketing site renders fully even if the backend is unreachable or unconfigured; only booking/admin features are affected
Usability	Guided, step-by-step booking flow; multilingual UI; clear calls-to-action throughout
Maintainability	TypeScript throughout; centralized content/data files (doctors, services, translations, images); documented in a developer guide
Scalability	Serverless backend (Supabase + Edge Functions) scales independently of the static frontend

10. Database Design

The backend schema (PostgreSQL, managed by Supabase) consists of six core tables:

Table	Purpose	Key Fields
doctors	Registered practitioners	id, name, title, is_active
services	Bookable consultation types	id, name, duration_minutes, is_active
appointments	Every patient booking	doctor_id, service_id, date, time, patient details, reason, status, payment info
reason_options	Configurable reason-for-visit checklist shown during booking	label, is_active
app_settings	Clinic-configurable settings	booking fee, UPI ID, payee name, WhatsApp number
blocked_slots	Staff-defined availability overrides	doctor_id, blocked_date, blocked_time (null = full day blocked)

appointments.status is constrained to pending, confirmed, cancelled, completed, or no-show. All tables are protected by Row-Level Security policies so that public (anonymous) access is limited to what the booking flow needs, while full read/write access is restricted to authenticated staff.

11. User Roles & Permissions

Role	Access
Visitor / Patient	Browse the public marketing site; submit a new appointment booking; no login required
Clinic Staff (Admin)	Log in at /admin/login; full access to Overview, Appointments, Availability, Payments, and Settings tabs; accounts provisioned manually by the practice, not self-serve

12. Third-Party Integrations

Service	Integration Purpose
Supabase	Database, authentication, and serverless email functions
Netlify	Hosting, CDN, and continuous deployment from Git
Unsplash	Supplemental stock photography for sections not using the clinic's own photos
WhatsApp (wa.me)	Click-to-chat link for patients to send payment confirmation screenshots
UPI (Unified Payments Interface)	QR-code-based manual payment collection at booking time

13. Deployment & Hosting

- **— Netlify, configured via netlify.toml (build command, publish directory, and a catch-all redirect so client-side routes like /admin work on refresh).**Hosting
- **— every push to the connected Git branch triggers an automatic build and deploy.**CI/CD
- **— Supabase URL and anon key are supplied as Netlify environment variables at build time.**Environment configuration
- **— the Netlify CLI can deploy a local production build directly for one-off releases.**Manual deploy option

14. Assumptions & Constraints

Assumptions

- A single doctor (Dr. Haritha Mandava) is the practitioner represented and scheduled at this time.
- Staff will verify UPI payments manually against the transaction ID and/or WhatsApp screenshot before marking a booking as Paid.
- The clinic has a working email address for the Supabase project to send transactional emails from.

Constraints

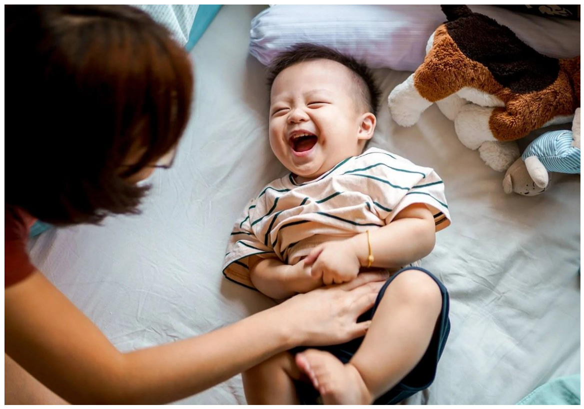
- No integrated payment gateway — all payment confirmation is manual/human-verified.
- The current data model scopes availability blocking to a single doctor; adding a second doctor requires extending the doctor picker and related booking/availability logic first.
- There is no automated test suite in place yet.

15. Risks & Mitigations

Risk	Mitigation
Manual payment verification is missed or delayed	Transaction ID is captured at booking and surfaced directly in the Appointments/Payments tabs for quick staff cross-checking
Backend misconfiguration breaks the whole site	isSupabaseConfigured flag ensures the marketing site still renders fully even if the backend is unavailable
Double-booking of a time slot	Booked, individually-blocked, and fully-blocked-day slots are all checked before a slot is offered
Admin account compromise	No public self-signup; accounts are provisioned manually by the practice in the Supabase dashboard

16. Future Enhancements

- Integrated online payment gateway for automatic payment capture and reconciliation.
- Support for multiple doctors and multi-branch scheduling.
- Automated SMS/WhatsApp appointment reminders ahead of the consultation.
- Patient-facing appointment history / self-reschedule and cancel.
- Automated test coverage (unit and end-to-end) using the already-installed Playwright toolchain.
- Analytics dashboard for booking trends and conversion tracking.



The warm, human moments the platform is built to represent for every family it serves.

17. Approval & Sign-off

By signing below, the stakeholders confirm that this Business Requirements Document accurately reflects the agreed scope, functionality, and technical approach for the BRD Hospital platform.

Name	Role	Signature	Date
Dr. Haritha Mandava	Project Sponsor / Client		
	Craftlanee — Development Partner		